Families of unexploitable agents that defect against themselves

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1 Introduction

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL. p means "the opponent cooperates with me" and q means "the opponent cooperates with DefectBot". FairBot is $\Box p$, graded FairBots are $F_k = \Box^k p$, graded PrudentBots are $P_j = \Box p \wedge \Box(\neg\Box^j\bot \to \neg q)$, and ConBot is $\neg\Box\bot$. An agent is *unexploitable* if no modal opponent ever gets it to cooperate while the opponent defects.

An earlier finding on this site described $W_n = \Box(\Box^n\bot \vee (p \wedge (q \vee \neg\Box q)))$: unexploitable, cooperative with FairBot and CooperateBot, yet defecting against its own copy. This finding describes four further families of unexploitable agents that defect against themselves, and how they relate.

Result

- **Gate families** $V_{n,r} = \Box(\Box^n\bot \vee (p \wedge (q \vee \neg\Box^r q)))$, which contain $W_n = V_{n,1}$. $V_{n,r}$ and F_k cooperate exactly when $k + r \leq n$; $V_{n,r}$ and P_j exactly when $r < n < j$; any two members defect against each other. For $r < n$ every member is unexploitable, and these members are pairwise distinct.

- **Prudent families** $G_{m,j} = \Box(\Box^m p \wedge (q \vee \Box(\Box^j\bot \vee \neg q)))$ **and** $H_{m,j} = \Box^{m+1}(p \wedge (q \vee \Box(\Box^j\bot \vee \neg q)))$, $m \geq j \geq 1$. Both are unexploitable, cooperate with FairBot and CooperateBot, defect against themselves, and cooperate with F_k exactly when $k \leq j$ and with P_i exactly when $i > m$. So $G_{m,j}$ and $H_{m,j}$ cannot be told apart by graded FairBots or PrudentBots; the agent $S_j = \Box^{j+1}(p \vee \neg\Box^{j+1} q)$ separates them ($H_{m,j}$ defects against it while it cooperates; against $G_{m,j}$ both defect). ConBot separates the two kinds of family: $G_{m,j}$ defects against it, every $V_{n,r}$ with $n \geq 1$ does not.

- **A common family** $J_{a,b,j} = \Box^{a+1}(\Box^b p \wedge (q \vee \Box(\Box^j\bot \vee \neg q)))$, which contains G ($a = 0$) and H ($b = 0$). Checked for $a, b \leq 5$, $j \leq 6$: $J_{a,b,j}$ cooperates with F_k exactly when $k \leq j$ and with P_i exactly when $i > a+b$,

and exact laws hold against a two-parameter family of separating probes; the 70 members with $j \leq a + b \leq 5$ are pairwise distinct. The size-13 member $J_{1,1,1}$ has the same outcomes as $G_{2,1}$ and $H_{2,1}$ against $F_1, \ldots, F_4$ and $P_0, \ldots, P_5$, yet two opponents of size at most 9 distinguish all three.

Why it matters

The unexploitable agents we found in the literature (FairBot, PrudentBot and its graded and probabilistic variants [1], [3]) all cooperate with their own copies. These families show that defecting against oneself is not a single odd example: it comes in several infinite families with different mechanisms (a delayed gate on the DefectBot test in V, a PrudentBot-style check in G, H and J). Their behaviour against graded FairBots and PrudentBots is described by simple thresholds, and these opponents alone do not identify an agent: distinct members can agree on all of them and need other probes to be told apart.

Proved vs checked

The laws for $V_{n,r}$, $G_{m,j}$ and $H_{m,j}$ (including the separations by ConBot and S_j) are proved for all indices in Lean 4, in the height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [4], cited, not formalised (see also [2]). The statements about $J_{a,b,j}$ and $J_{1,1,1}$ are checked, not proved: exact tables for the stated parameter ranges, with every member of size at most 15 independently certified by GL fixed-point solving. The unexploitability of $J_{a,b,j}$ is supported by a short argument (the agent implies $\Box^{a+b+1}p$) and randomised safety tests, not by a proof for all indices.

Novelty statement

We believe this result is new. Our prior-art searches (2026-10-17 to 2026-10-18) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: named agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10; obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem.
- LessWrong post on Barasz et al. and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling"; Garrabrant, "Modal SAT: Introduction" (2015).

- Mennen, “Modal Combat for games other than the prisoner’s dilemma”.
- Critch, Parametric Bounded Löb’s Theorem and Robust Cooperation of Bounded Agents (arXiv 1602.04184), first sections.
- Searches for round-robin, tournament, replicator or ESS analyses of modal agents: only iterated-PD and FDT evolution papers (e.g. arXiv 2005.05154, no modal agents).
- github.com/machine-intelligence/provability (owner, 2026-10-18): its example agent file (target.ua) and the Haskell agent catalogue (Agents.hs) (CB, DB, FB, FB_1..5, TbF_1..3, FB_r, MB, WB, WB_2, aMB, sMB, IB, PB, PB_2, PB_3, NB, JB, TB, QB_1..3, modal UDT). Evaluated in height semantics (our reading of its [k] and boxk notation as PA+k provability): none of them defects against itself while cooperating with FairBot and CooperateBot; none has the form of V, G, H or J.
- Barasz et al. 2014 (arXiv 1401.5577), full text; LaVictoire et al. 2014 (Program Equilibrium via Löb’s Theorem).
- LessWrong post on Barasz et al. and comments; Arbital “Modal combat”.
- Garrabrant, “Prisoners’ Dilemma with Costs to Modeling” and “Modal SAT: Introduction”; Mennen, “Modal Combat for games other than the prisoner’s dilemma”.
- Critch, Parametric Bounded Löb (arXiv 1602.04184), first sections.
- Oesterheld 2022, A Note on the Compatibility of Different Robust Program Equilibria of the Prisoner’s Dilemma (arXiv 2211.05057).

We found no earlier statement of it, but we can’t be certain. If you know of prior work, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was partly checked by scripts that run Lean proofs (claims K14, K18, K21) and partly checked by script (claims K20, K22, K23); the proof and check files are included so anyone can re-run them.

2 Claims

Claim K14

`For all n>=0, r>=1, k>=1, j>=0 (height semantics of closed GL fixed points), V_{n,r}=[]([]^nF|(p&(q|~[]^r q))): CC with F_k iff k+r<=n; with P_j iff r<n<j; any two V’s defect. For r<n each is universally unexploitable, all pairwise distinct. Lean.`

Status: derived ů author check only.

Verified.

- `checks/K14-c046_vfam_lean.py`: embedded Lean 4 theorems qclause_r, v_unexploitable, v_db, v_cb, v_fk, pj_db, v_pj, v_vm, v_params compile with no sorryAx; 5 false variants rejected; literal height tables n<=10, r<=5, k,j<=12 and V-vs-V n,m<=6, r,s<=3 agree; 35 r<n agents have distinct F/P outcome vectors; size(V_{n,r})=n+r+9 — pass (Lean-proved, no sorryAx)
- `checks/K14-c046_vfam_lean.py`: embedded Lean 4 theorems qclause_r, v_unexploitable, v_db, v_cb, v_fk, pj_db, v_pj, v_vm, v_params compile with no sorryAx; 5 false variants rejected; literal height tables agree; 35 r<n agents have distinct F/P outcome vectors — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Closed GL fixed points are letter-free and decided by height on finite linear frames (Boolos, Logic of Provability ch. 7); v_unexploitable quantifies over arbitrary opponent traces, hence all modal opponents.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K18

`For all m>=j>=1 (height semantics of closed GL fixed points), G_{m,j}=[]([]^m p&(q|[]([]^jF|~q))), size m+j+10, is unexploitable, coops with FB and CB, defects vs itself; CC with F_k iff k<=j, P_i iff i>m; defects vs ~[]F unlike every V_{n,r} with n>=1. Lean.`

Status: derived ů author check only.

Verified.

- `checks/K18-c046_gfam_lean.py`: embedded Lean 4 theorems gq, g_unexploitable, g_db, g_cb, g_conbot, g_fk, g_pj, g_gg, v_conbot, g_params compile with no sorryAx; 4 false variants rejected; literal height tables m<=8, j<=9, k,i<=11 and G-vs-G m,m'<=5 agree; independent string solver: self/FB/CB/DB/ConBot for m,j<=3, W_2 coops with ConBot; size(G_{m,j})=m+j+10 — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Closed GL fixed points are letter-free and decided by height on finite linear frames (Boolos, Logic of Provability ch. 7); g_unexploitable quantifies over arbitrary opponent traces, hence all modal opponents.
- reasoning: Scope fix only: the Lean lemma v_conbot already assumes n\>=1; V_{0,1} defects against ConBot (the skeptic, k18_conbot_scope_failure.py).

- reasoning: Scope corrected: v_conbot (Lean, in c046_gfam_lean.py) assumes 1\<=n; V_{0,r} defects vs ConBot (the skeptic's k18_conbot_scope_failure.py).

Prior art. Possibly new: no prior art found in the owner's search.

Claim K20

`For all H_{m,j}=box^{m+1}(p&(q|box(box^j F|~q))), 1<=j<=m, size m+j+10<=15 (six agents), self-play is DD, FB/CB CC; CC with F_k iff k<=j (k<=4), P_i iff i>m (i<=5). GLS certifies 102 cells including separators from matching G and bounded V.`

Status: derived ů author check only.

Verified.

- `checks/K20-outer_prudent_family.py`: Exhausts six H parameters of size<=15; all 102 pinned fixed-point cells pass GLS with mutation controls; common F/P rows distinguish all six; explicit matching-G and ConBot/V separators. — pass

Evidence.

- reasoning: H implies box^(m+1)p; eventual cooperation by H forces p at every fixed height and hence prevents exploitation by any modal opponent.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K21

`For m>=1, j>=1 (height semantics of closed GL fixed points), H_{m,j}=[]^{m+1}(p&(q|[]([]^jF| is unexploitable; DB trace h<=m; CC with F_k iff k<=j, P_i iff i>m; H vs H defect (j<=m); S_j=[]^{j+1}(p|~[]^{j+1}q) gives D/C vs H_{m,j}, D/D vs G_{m,j}. Lean.`

Status: derived ů author check only.

Verified.

- `checks/K21-c046_hfam_lean.py`: embedded Lean 4 theorems h_fail, h_unexploitable, h_db, h_cb, h_conbot, h_fk, h_pj, h_hh, s_db, h_sep, g_sep, h_params compile with no sorryAx; 4 false variants rejected; literal height tables m,j<=8, k,i<=11, H-vs-H m,m'<=5, H/G vs S_j agree; independent string solver for m,j<=3: self, FB, CB, DB, ConBot, F_1..4, P_0..5, S_j; size(H)=m+j+10, size(S_j)=2j+6 — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Closed GL fixed points are letter-free and decided by height on finite linear frames (Boolos, Logic of Provability ch. 7); h_unexploitable quantifies over arbitrary opponent traces, hence all modal opponents.
- reasoning: Scope narrowed to m\>=1; Lean theorem g_sep assumes 1\<=m, and at m=0 G=H syntactically.
- reasoning: Scope fix: g_sep (Lean) assumes 1\<=m; at m=0 G_{0,j}=H_{0,j} (another researcher's k21_zero_scope.py).

Prior art. Possibly new: no prior art found in the owner's search.

Claim K22

`In L, the size-13 agent J=[][]([]p&(q|[]([]F|~q))) has self DD and FB/CB CC. Its F1..4/P0..5 outcomes match G21 and H21, but two size<=9 opponents distinguish all three; all 48 stated cells are GLS-certified.`
Status: derived ů author check only.

Verified.

- `checks/K22-mixed_prudent.py`: 48 hash-pinned GLS-certified cells with mutation controls; all three agents size13; named rows agree; two small opponents distinguish G21,J,H21. — pass

Evidence.

- reasoning: J implies box^3 p, so eventual J cooperation forces p at every fixed height, precluding exploitation by any modal opponent.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K23

For $J_{a,b,j} = \Box^{a+1}(\Box^b p \wedge (q \vee \Box(\Box^j F \vee \neg q)))$, $0 \le a, b \le 5$, $1 \le j \le 6$: CC with F_k ($k \le 9$) iff $k \le j$, P_i ($i \le 9$) iff $i > a + b$; vs $S_{u,t}$ ($u, t \le 8$): CC if $u \le j$, else D/C iff $t + b \le u$, else DD. The 70 with $j \le a + b \le 5$ are distinct.
Status: derived ů author check only.

Verified.

- `checks/K23-c046_jfam.py`: literal tables a,b<=5, j<=6: DB/CB/ConBot, F_k (k<=9), P_i (i<=9), S_{u,t} (u,t<=8) laws incl. defection heights, J vs J defect, random-trace safety; 70 members (j<=a+b<=5) have distinct vectors; independent string solver: 882 cells for all 21 members of size<=15 — pass

- `checks/K23-c046_jfam_gls.py`: all 21 J_{a,b,j} with 1<=j<=a+b, size<=15: 265 cells (self, CB, DB, ConBot, F_j/F_{j+1}, P_{a+b}/P_{a+b+1}, S_{j,j+3}, S_{u,t} boundaries t+b=u vs u+1 for u=j+1,j+2, G21/J111/H21 split on S_{2,1},S_{2,2}) match the K23 law formulas and pass GLS fixed-point certificates with flipped-candidate rejection — pass

Evidence.

- reasoning: S_{u,t} has DB trace h\<=u-1, so for u\>j J's q-test fails from u and J coops exactly at h\<=u+a; S needs J at all g\<=a+b+t, satisfied iff t+b\<=u.
- reasoning: another researcher's `placement_certificates.py` (a check not published with this finding): 160 GLS-certified cells for all 8 triples with 1\<=j\<=a+b\<=2 vs F_1..3, P_0..3, S_{u,t} (u,t\<=3), self, CB, DB, ConBot agree with these laws; another researcher's independent height solver reproduces the pinned rows (sha c0422274...).

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261017-1126`, claims K14, K18, K20, K21, K22, K23, run-folder commit `662166ecff0651902cc5fcb04bf747c62d795af2`, published 2026-10-18.

3 How to verify

Each finding has its check scripts in `claims/gate-families/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

4 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] George Boolos. *The Logic of Provability.* Cambridge University Press, 1993.

[3] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[4] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.